**Ministry of Higher Education**

**Thebes Academy**

**Thebes Higher Institute for Computer and Management Sciences**

# School Management System

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# Abstract

We are developing a comprehensive School Management System that digitizes and automates administrative and academic processes for educational institutions. Our platform provides a centralized solution for managing student data, attendance records, grades, fee payments, and stakeholder communication in one unified system. For teachers, the platform offers an all-in-one dashboard to record attendance, upload grades, share learning materials, assign homework, and communicate directly with students and parents. Students can access their timetable, view grades and attendance, submit homework, and download educational materials from a single dashboard. Parents receive real-time updates about their child's academic progress, attendance records, and school announcements through a dedicated parent portal. The system features AI-powered analytics that identify weak students, predict performance trends, and recommend personalized improvement strategies. Administrators benefit from simple visual dashboards with role-based access control. By combining these features, our platform reduces administrative workload, improves data accuracy, enhances parent engagement, and modernizes school operations for the digital age.

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# Acknowledgments

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# Chapter One

# Introduction

# Chapter 1: Introduction

## 1.1-Problem Definition:

Many schools still rely on manual systems using paper records, Excel sheets, and WhatsApp communication for daily operations. This fragmented approach leads to several critical issues that affect the entire school community. Manual data management is highly prone to errors, with misplaced records and incorrect data entry causing inaccurate student information. The lack of a centralized platform means student data is scattered across different systems, making it difficult to get a complete picture of student performance. Tracking fees becomes challenging without a unified system, leading to delayed payments and accounting errors. Monitoring student performance requires manually compiling data from multiple sources, making it hard to identify struggling students early. Communication between administrators, teachers, students, and parents remains weak and inefficient through informal channels. There is a clear need for a centralized digital solution that addresses these problems.

No centralized system for managing student data; schools store information across different platforms and paper records. Manual attendance tracking is time-consuming and prone to errors and data loss. Difficulty tracking fee payments and managing financial records leads to accounting inconsistencies. No unified platform for homework submission, grade viewing, and learning material access. Teachers must use multiple tools to upload assignments, record grades, and communicate with parents. Lack of data analytics to identify weak students or predict academic performance

## 1.2 Problems Solution:

1. Centralized database that stores all student information, grades, attendance, and fees in one secure platform.
2. Role-based dashboards for Admin, Teacher, Student, and Parent with user-friendly interfaces.
3. Automated attendance tracking with digital recording and instant report generation.
4. All-in-one teacher platform to record attendance, upload grades, share materials, and assign homework.
5. AI-powered analytics to analyze student data, identify weak students, predict performance, and recommend improvement strategies.

## 1.3 Aim and Objectives:

1. Automate administrative processes to reduce workload on school staff and minimize errors.
2. Provide real-time communication channels between all stakeholders (Admin, Teacher, Student, Parent).
3. Improve data accuracy and transparency in student records, grades, and fee tracking.
4. Create a centralized platform where students can view timetable, grades, attendance, submit homework, and download materials.
5. Enable teachers to manage all classroom activities from a single dashboard.
6. Integrate AI analytics for predicting student performance and identifying at-risk students early.

7. Support both web and mobile platforms for accessibility anywhere, anytime.
8. Provide simple visual dashboards with local customization for Arabic schools.

## 1.4 Challenges:

1. Centralized database that stores all student information, grades, attendance, and fees in one secure platform.
2. Role-based dashboards for Admin, Teacher, Student, and Parent with user-friendly interfaces.
3. Automated attendance tracking with digital recording and instant report generation.
4. Integrated online payment system supporting Fawry and Vodafone Cash for fee collection.
5. Real-time messaging system between teachers, students, and parents with notifications.
6. All-in-one teacher platform to record attendance, upload grades, share materials, and assign homework.
7. AI-powered analytics to analyze student data, identify weak students, predict performance, and recommend improvement strategies.

# TimeLine:

| Time | System flow | Description |
|---|---|---|
| Oct 2025: Nov 2025 | Planning | Planning and Requirement Analysis Defining Requirements |
| Nov 2025: Dec 2025 | Designing | Defines all the architectural modules Designing the web architecture |
| Dec 2025: Jan 2026 | Break | First Semester Examinations |
| Jan 2026: March 2026 | Implementation | Building or developing the web application |
| March 2026: Apr 2026 | Testing | Testing the web application |
| Apr 2026: May 2026 | Deployment | Deployment and Maintenance then based on the feedback the web application may be released |

# Chapter Two

# Literature Review

# Chapter 2: Literature Review

## 2.1 PowerSchool:

### ❍ Defects:

1. Complex and difficult-to-use admin panel requiring technical expertise to navigate.
2. Homework and learning materials are not fully integrated for students in one place.
3. Teachers must use multiple tools to upload assignments, materials, and communicate with students.
4. Analytics and reports are powerful but complicated and not localized for Arabic schools.
5. No built-in AI for predicting student performance or identifying weak students.
6. Expensive and over-engineered for small to medium-sized schools.
7. Recent security/data breach raised concerns about data protection.

### ❍ Features:

1. Centralized system for student data, grades, attendance, and reports.
2. Parent and student portals for real-time academic updates.
3. Widely used and supports integrations with external educational platforms.
4. Provides reporting and administrative tools for comprehensive school management.

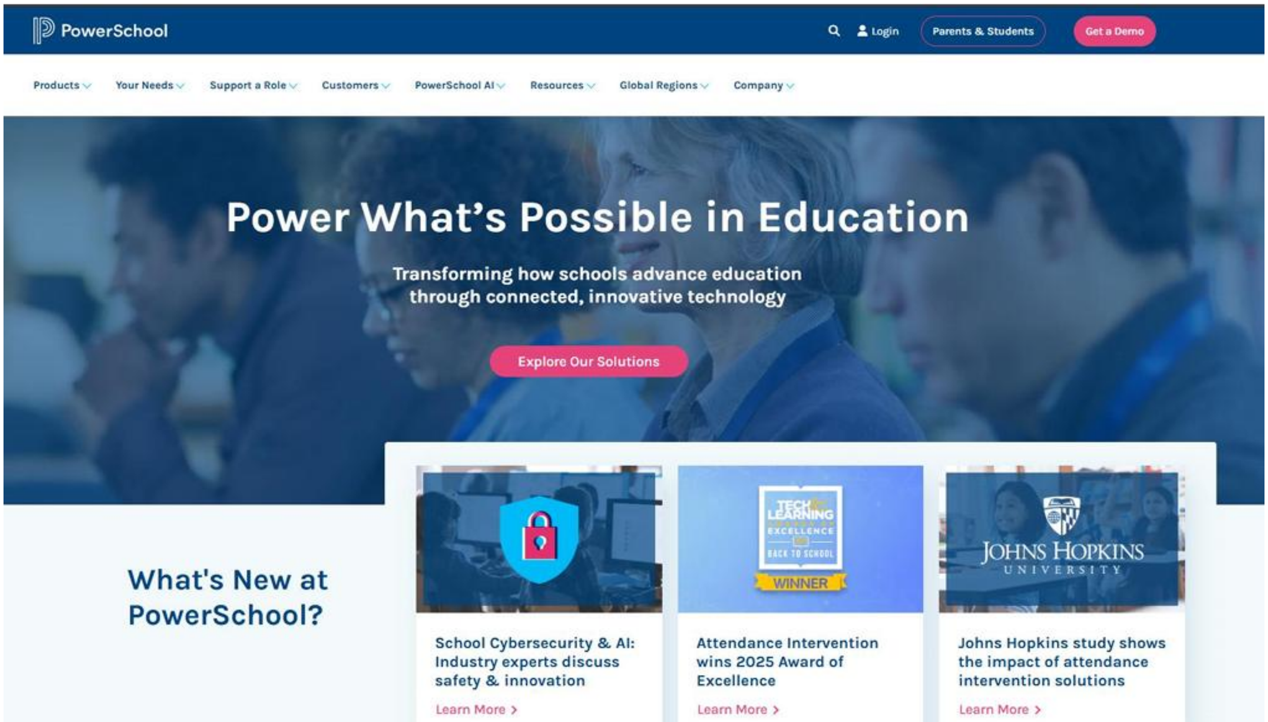


Figure 2.1 **PowerSchool** [Reference 1]

## 2.2 openSIS (Open-Source/Cloud-Based School Management System):

### Defects:

1. Requires technical skills – Open-source version needs internal IT developers for setup, updates, and customization.
2. Limited features in free version – Advanced features like analytics, online payments, or communication tools are only in paid versions.
3. User interface not modern or intuitive – Compared to newer systems, the UI may feel outdated and harder for teachers and parents to use.

4. No built-in AI or predictive analytics – It only records grades and attendance but doesn't suggest improvements or predict student performance.
5. No integrated messaging system for communication between stakeholders

## Features:

1. Open-source and customizable – Schools can fully modify features and interface based on their own needs without licensing restrictions.
2. Cost-effective – Free community edition and lower-cost cloud packages reduce financial barriers for schools.
3. Good documentation and active community – Developers and IT teams can access guides, forums, and community support easily.
4. Modular system – Supports basic student management, attendance, scheduling, grades, and reports.

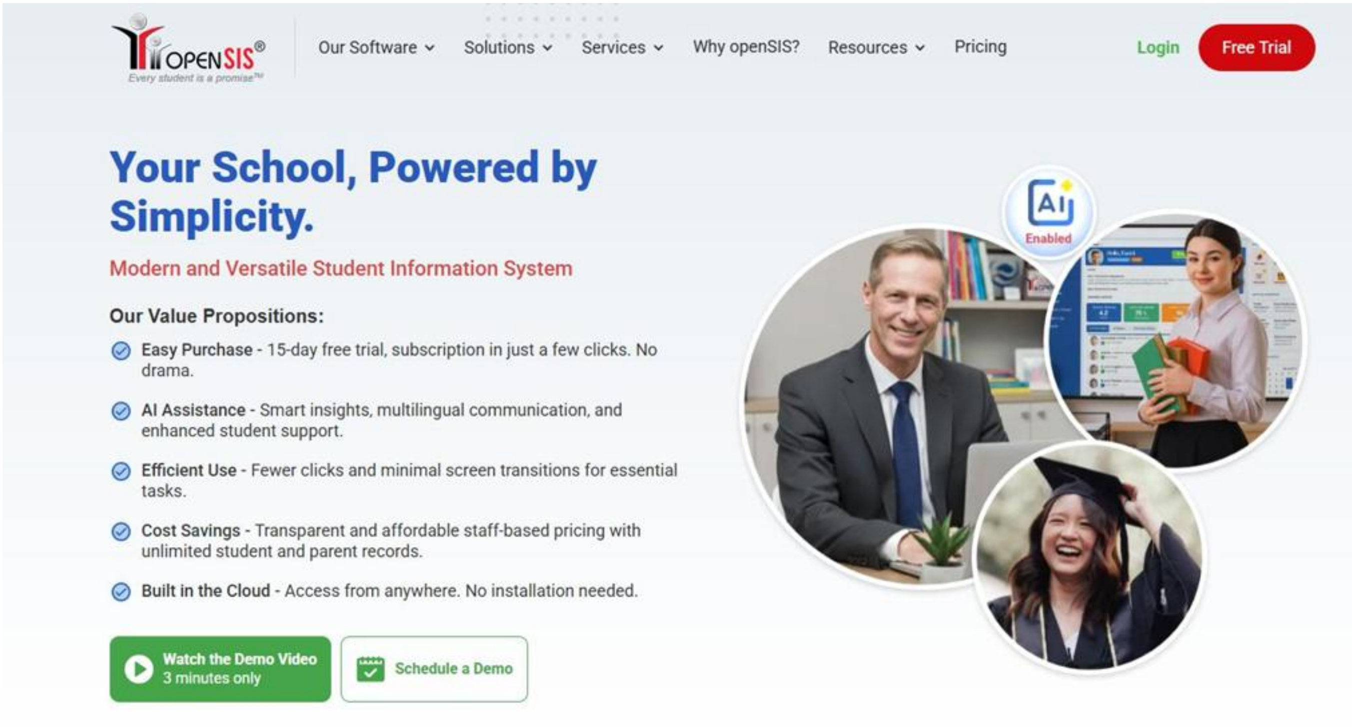


Figure 2.2 Ma3ak **openSIS** [Reference 2]

## 2.4 How Our System Solves These Problems:

### ➢ Solutions:

1. **Simple and User-Friendly Dashboard:** We provide a simple and user-friendly dashboard with role-based access control (Admin, Teacher, Student, Parent) that can be managed without IT specialists.
2. **All-in-One Student Platform:** Our system allows students to view timetable, grades, attendance, submit homework, and download materials from a single dashboard.
3. **Integrated Teacher Tools:** Teachers have an all-in-one platform to record attendance, upload grades, share materials, assign homework, and chat directly with students and parents.
4. **AI Analytics:** Our system uses AI to analyze student data, identify weak students, predict performance, and recommend improvement strategies.

5. **Cloud-Based Setup:** Fully cloud-based system with easy setup; no need for programming knowledge.
6. **Modern UI:** Modern, user-friendly interface with separate dashboards for Admin, Teacher, Student, and Parent.
7. **Digital Attendance:** Digital attendance recording, automated grade entry, and instant report generation.

# Chapter Three
# System Design

# Chapter 3: System Design

## Overview:

**3.1 Use Case Diagram**

**Description:**

**The use case diagram represents the interactions between the system's actors (Admin, Teacher, Student, Parent) and the key functionalities provided by the School Management System.**

**Key Features:**

- **Visualizes main user actions such as registration, authentication, academic activities, communication, and financial transactions.**
- **Actors: Teacher, Parent, Student, and Admin.**

**3.2 Activity Diagram**

**Description:**

**The activity diagram models the workflow of the School Management System, showing the sequence of operations and decision points for key processes across different user roles.**

**Key Features:**

- **Displays parallel and sequential activities for Admin, Teacher, Student, and Parent.**
- **Highlights decision points based on user type selection.**

**3.3 Sequence Diagram**

**Description:**

**The sequence diagram details the order of interactions between objects in the system for specific, critical functionalities over time. It shows message exchange between components like User Interface, Controllers, and Database.**

**Key Features:**

- **Focuses on the dynamic behavior of the system**
- **Tracks message exchanges between components**
- **Highlights real-time communication and conditional steps**

### 3.4 Class Diagram

**Description:**

**The class diagram defines the system's structure by outlining the system's classes, their attributes, and relationships. It shows inheritance hierarchies and associations between different user types and system components.**

**Key Features:**

- **Shows relationships like inheritance and association between classes**
- **Defines classes like User, Teacher, Student, Parent, and Admin**
- **Captures core entities such as assignments, payments, messages, and exams**

### 3.5 Entity-Relationship Diagram (ERD)

**Description:**

**The ERD illustrates the logical database design of the School Management System. It defines entities, attributes, and relationships among data tables.**

**Key Features:**

- **Captures relationships among core entities including Users, Students, Teachers, Parents, and Academic records**
- **Enables tracking of attendance, grades, AI recommendations, and user status**

- **Defines primary keys, foreign keys, and status flags for real-time monitoring**

## 3.1 Use Case Diagram:

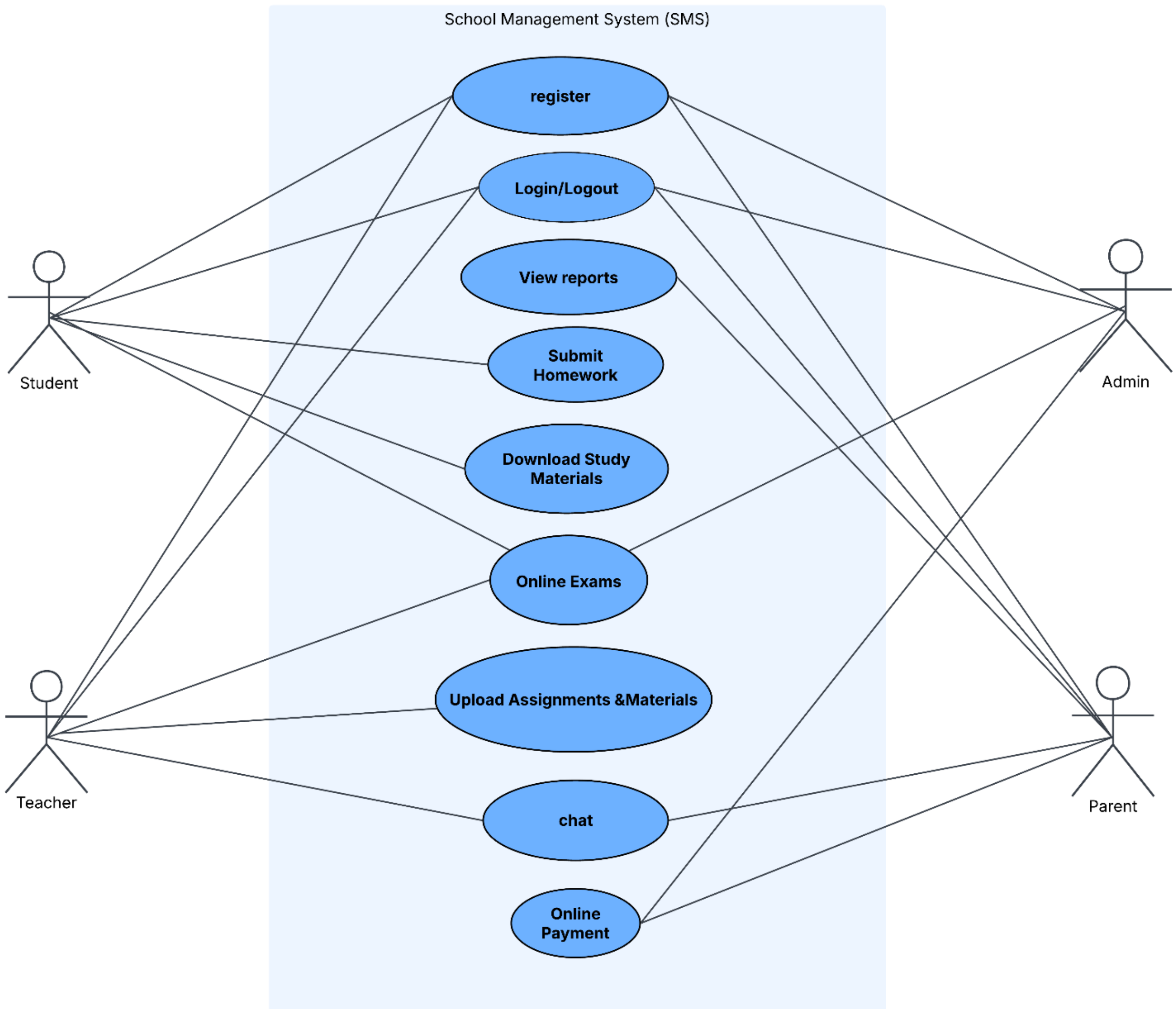


Figure 3.1Use Case

## 3.2 Activity Diagram:

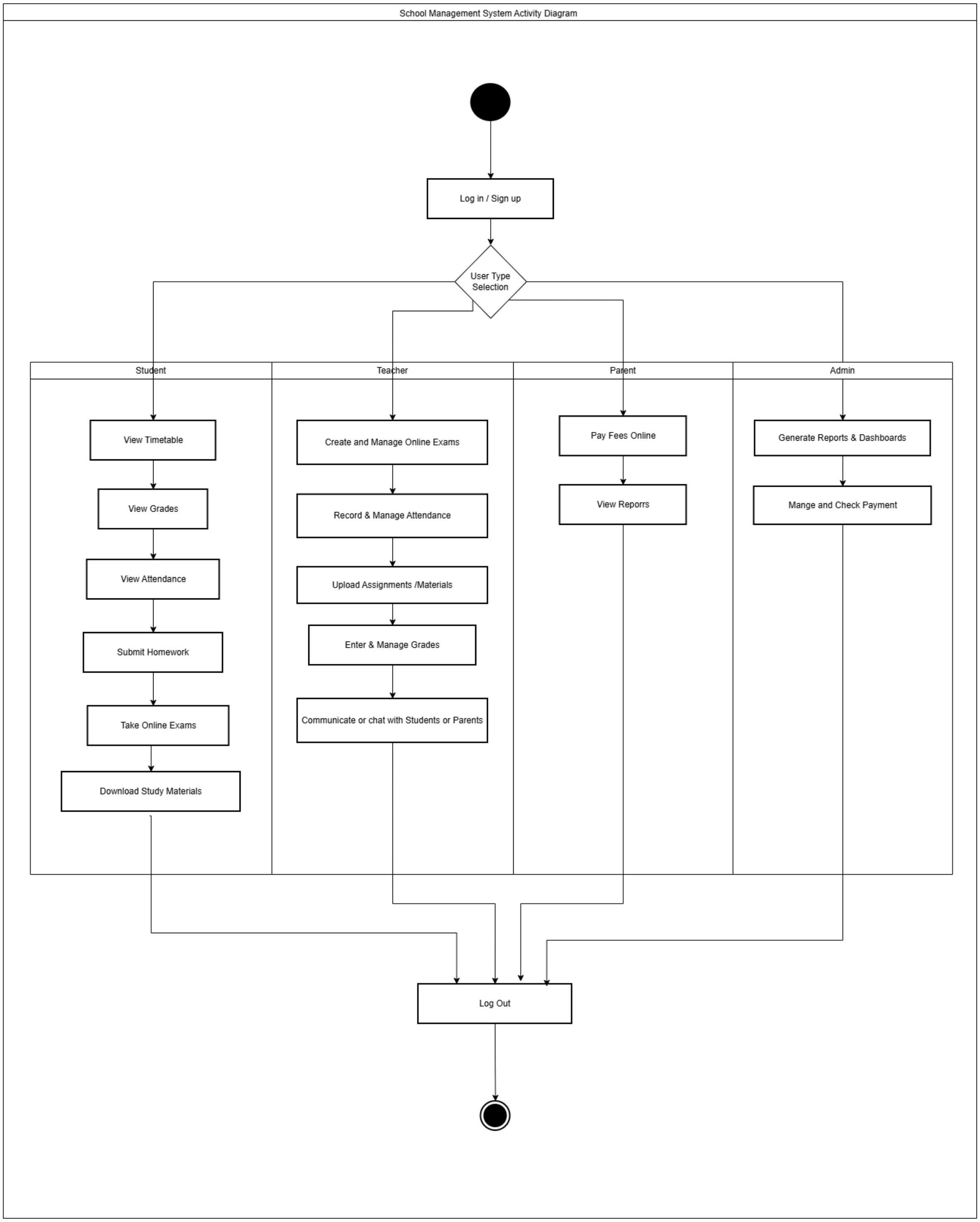


Figure 3.2 Activity Diagram

## 3.3 Sequence Diagram:

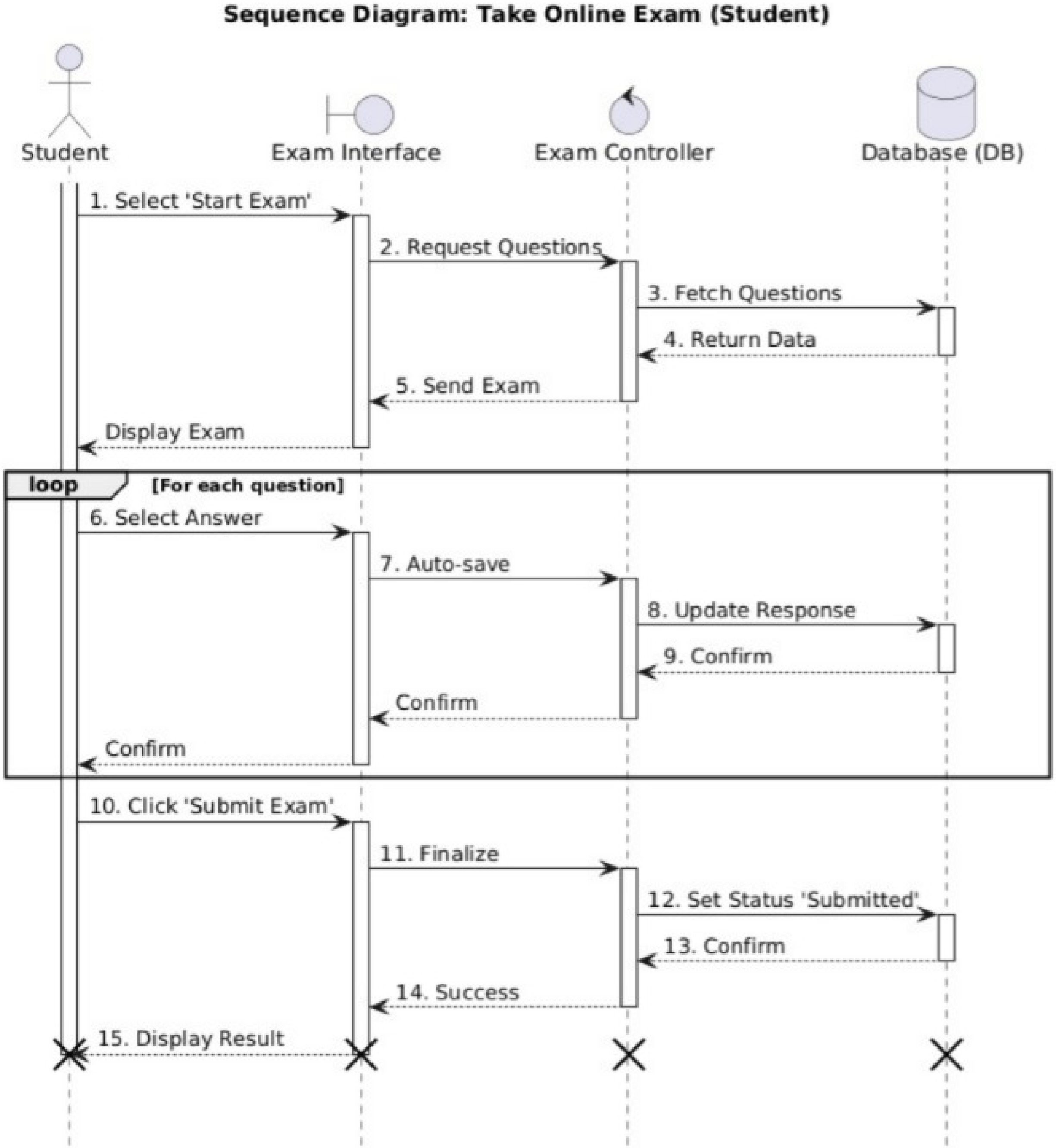


Figure 3.3 Sequence Diagram 1

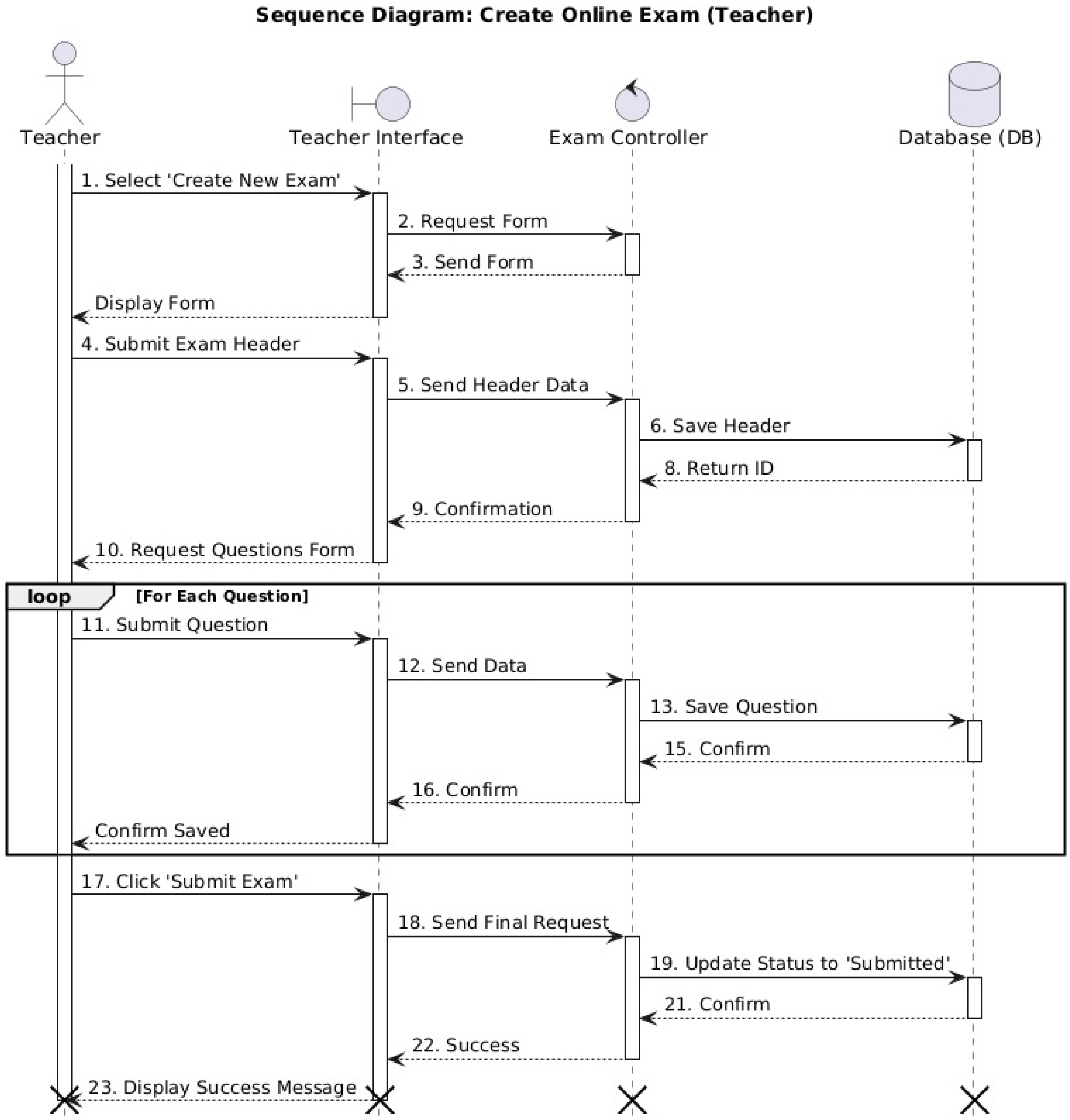


Figure 3.4 Sequence Diagram 2

## 3.6 Class Diagram:

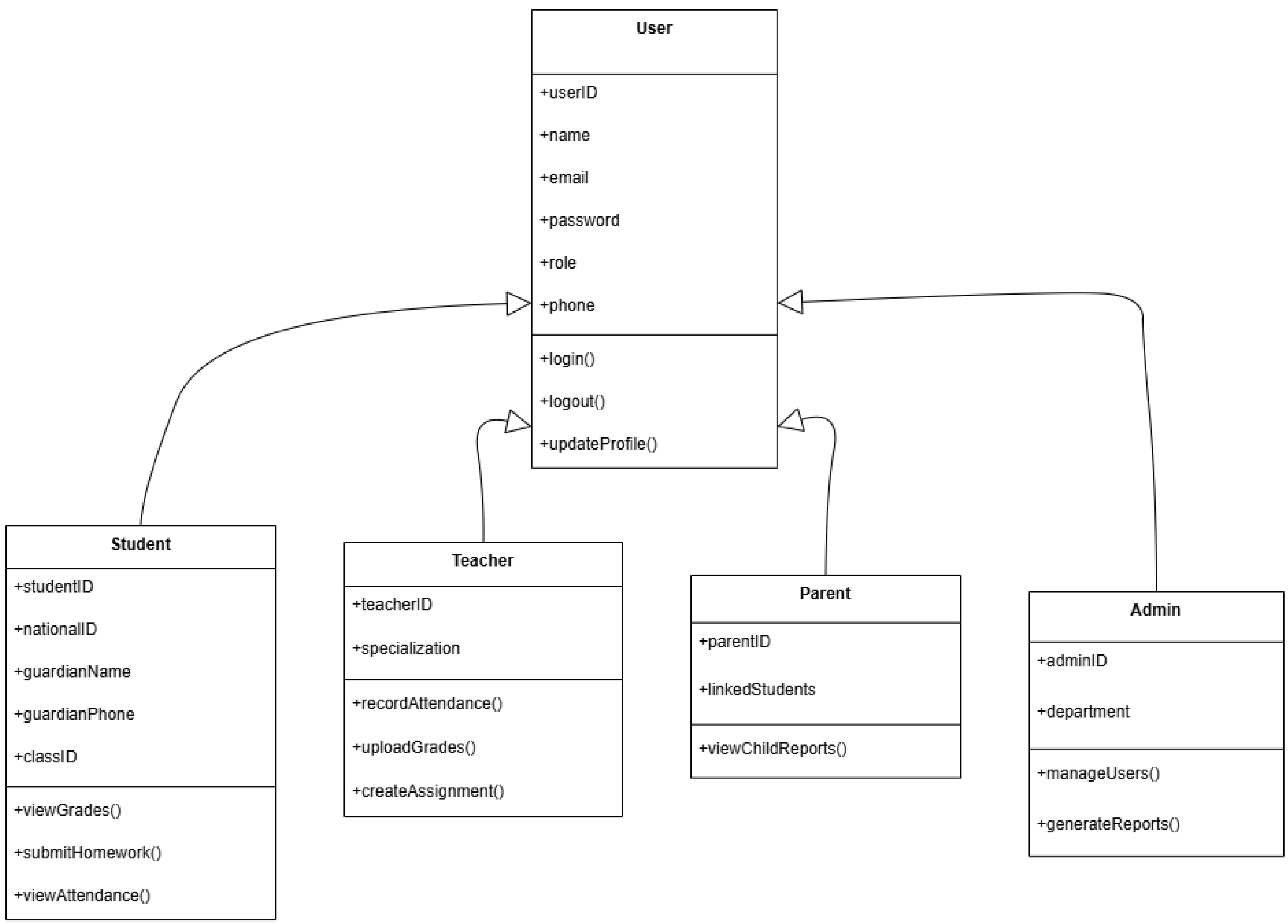


Figure 3.5 Class Diagram

## 3.7 ERD:

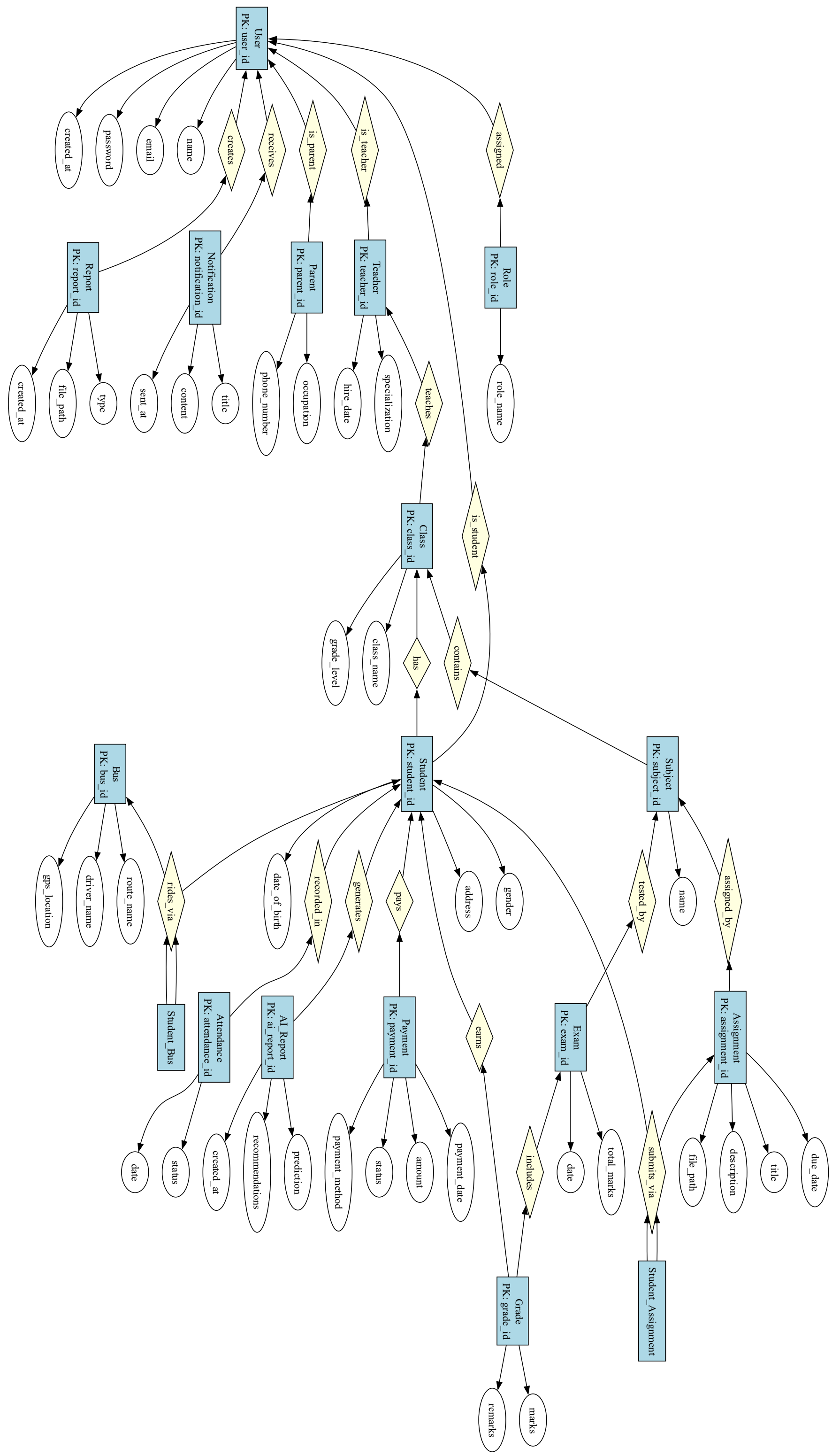


Figure 3.6 ERD

# References List

## References List


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